

Mathematical proof - Set 2 - Solutions

Specification Prf1-Prf4.

Answer key

Question	Answer	Topic
1	D	Mathematical proof
2	C	Mathematical proof
3	A	Mathematical proof
4	A	Mathematical proof
5	A	Mathematical proof
6	A	Mathematical proof
7	F	Mathematical proof
8	E	Mathematical proof
9	C	Mathematical proof

Worked solutions

Question 1 PRF-05 · Mathematical proof · spec Prf3

The sums 1 , $1 + 3$, $1 + 3 + 5$, $1 + 3 + 5 + 7$ are 1 , 4 , 9 , 16 . A student conjectures that the sum of the first n odd numbers is n^2 . Which of the following is a valid justification?

Answer: D

A applies the arithmetic-series formula to all n . (C also gives a correct calculation but B, D and E are not justifications.)

Full method - how and why

Distinguishing a justification from a pattern-spot. The valid answer must establish the formula for all n .

- 1 The first n odd numbers are $1, 3, 5, \dots$ - an arithmetic series with $a = 1$, $d = 2$.

why: Recognising the structure is what lets a general formula apply.

- 2 Apply $S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(2 + 2(n - 1))$.

why: The standard arithmetic-series sum, valid for every n .

- 3 Simplify: $\frac{n}{2} \times 2n = n^2$.

why: $2 + 2n - 2 = 2n$.

- 4 That holds for all n , so it justifies the conjecture.

why: By contrast, 'the pattern continues with 25' checks one more case, and 'four cases is enough' explicitly is not a proof.

Where it goes wrong

- Accepting 'four cases have been checked' - the whole point of this topic is that it is not enough.
- Accepting ' n^2 is odd when n is odd' - a true but completely irrelevant statement.
- Being drawn to the even-numbers argument; its calculation happens to work out, but it is presented with an unjustified aside, whereas the series argument is clean and standard.

Takeaway A justification must apply to every n ; extending a pattern one more term never does.

Question 2 PRF-06 · Mathematical proof · spec Prf1

Which of the following statements can be disproved by a single counterexample?

- I Every prime number greater than 2 is odd.
- II Every integer of the form $4k + 3$ is prime.
- III There exists an even prime number.

Answer: C

Only a 'for all' statement can be disproved by one counterexample. I is true (no counterexample exists). II is false and $k = 3$ (15) disproves it. III is an existence statement - a counterexample is meaningless; it is proved by the example 2.

Full method - how and why

Only a **universal** statement that is **false** can be disproved by a counterexample. Two separate tests.

- 1 I: 'every prime > 2 is odd' is universal - but it is **true** (any even number > 2 has factor 2), so no counterexample exists.
why: You cannot disprove a true statement.
- 2 II: 'every integer $4k + 3$ is prime' is universal and **false**: $k = 3$ gives $15 = 3 \times 5$. Disprovable. **Yes.**
why: One failing case is a counterexample.
- 3 III: 'there exists an even prime' is **existential**, not universal - and it is true (2). A counterexample is not even the right tool.
why: To disprove an existential statement you would have to rule out every case, which is not a 'single counterexample'.
- 4 So II only.
why: Match to the list.

Where it goes wrong

- Including I because it is a 'for all' statement - the statement also has to be false.
- Including III by confusing what disproves an existential claim.

Takeaway Counterexamples disprove false universal statements only. Existential statements are proved by example and disproved only by a general argument.

Question 3 PRF-09 · Mathematical proof · spec Prf1

Here is a proof that there are infinitely many primes, with the steps out of order.

- I Then N leaves remainder 1 on division by each of $p_1, \dots, p_n$, so no p_i divides N .
- II Suppose there are only finitely many primes $p_1, p_2, \dots, p_n$, and let $N = p_1 p_2 \cdots p_n + 1$.
- III So N has a prime factor that is not in the list, which is a contradiction.
- IV But $N > 1$, so N has at least one prime factor.

Which is the correct order?

Answer: A

Define N (II), show no listed prime divides it (I), note that N nevertheless has some prime factor (IV), and conclude that this factor is a new prime (III). The conclusion III cannot come before both I and IV have been established.

Full method - how and why

Ordering Euclid's proof. Again the skeleton is assume-deduce-contradict, and the dependencies fix the order.

- 1 II first: assume finitely many primes $p_1, \dots, p_n$ and construct $N = p_1 p_2 \cdots p_n + 1$.
why: Everything else refers to N , so it must be defined first.
- 2 Then I: each p_i divides $p_1 \cdots p_n$, so dividing N by p_i leaves remainder 1 - no listed prime divides N .
why: This is the key property of the $+1$.
- 3 Then IV: $N > 1$, so it has at least one prime factor.
why: Every integer greater than 1 has a prime factor - a separate fact, independent of I.
- 4 Finally III: that prime factor is not in the list, contradicting the assumption that the list was complete. Order: II, I, IV, III.
why: The contradiction needs **both** I (none of the listed primes works) and IV (some prime does).

Where it goes wrong

- Putting III immediately after II, before either supporting fact has been established.
- Thinking the proof claims N is prime - it need not be; it merely has a prime factor outside the list.

Takeaway In a contradiction proof, the contradiction is the last line, and it must rest on everything before it.

Question 4 PRF-10 · Mathematical proof · spec Prf1

Consider the claim: **if x and y are real numbers with $x + y > 10$, then $x > 5$ or $y > 5$** . Which of the following proves it?

Answer: A

A is proof by contrapositive/contradiction. B assumes the conclusion, C is an example, D proves something false, E's conclusion ('both exceed 5') is wrong.

Full method - how and why

Proving 'if P then $(A \text{ or } B)$ '. Assuming both A and B fail and deriving a contradiction is the natural route.

- 1 Negate the conclusion: 'not $(x > 5 \text{ or } y > 5)$ ' means $x \leq 5$ **and** $y \leq 5$ (De Morgan).

why: Negating an 'or' gives an 'and' - and it gives you two usable facts instead of a disjunction.

- 2 Add the inequalities: $x + y \leq 10$.

why: Inequalities of the same direction may be added.

- 3 That contradicts the hypothesis $x + y > 10$, so the conclusion must hold.

why: Hypothesis true and conclusion false is impossible, which is exactly what proves the implication.

Where it goes wrong

- 'If $x > 5$ then the conclusion holds' - this assumes part of what is to be proved.
- Testing $x = y = 6$ - a single example proves nothing about all x, y .
- 'Suppose $x > 5$, then $y > 10 - x$, so $y > 5$ ' - the algebra is wrong and it proves a false statement (both exceeding 5).
- 'The average exceeds 5 so both exceed 5' - the average exceeding 5 only forces **at least one** to.

Takeaway To prove an 'or', negate it into an 'and' and look for a contradiction - De Morgan turns a weak conclusion into strong assumptions.

Question 5 PRF-11 · Mathematical proof · spec Prf3

A student notices that $1 + 2 = 3$, $4 + 5 + 6 = 7 + 8$, $9 + 10 + 11 + 12 = 13 + 14 + 15$ and conjectures that the pattern continues. Which of the following describes the n th line of the pattern?

Answer: A

Line 1 starts at 1, line 2 at 4, line 3 at 9: start n^2 . Left sides have 2, 3, 4 numbers ($n + 1$), right sides 1, 2, 3 (n).

Full method - how and why

Describing a pattern precisely: check the start value and the two counts separately against every given line.

- 1 Starting values: 1, 4, 9 for lines 1, 2, 3 - these are n^2 .

why: Test against all three lines, not just one.

- 2 Count the left sides: $\{1, 2\}$, $\{4, 5, 6\}$, $\{9, 10, 11, 12\}$ - that is 2, 3, 4 terms, i.e. $n + 1$.

why: Count carefully; being off by one here is the whole point of the question.

- 3 Count the right sides: $\{3\}$, $\{7, 8\}$, $\{13, 14, 15\}$ - that is 1, 2, 3 terms, i.e. n .

why: Left has one more term than right, which is what makes the equality possible (the right-hand terms are larger).

- 4 So: starts at n^2 , with $n + 1$ terms on the left and n on the right.

why: Check the total count: $(n + 1) + n = 2n + 1$, and indeed line 3 runs from 9 to 15, seven numbers ✓.

Where it goes wrong

- Swapping the counts (n left, $n + 1$ right) - the sides would not balance, since the right-hand numbers are bigger.
- Reading the start as $2n - 1$ or $n(n + 1)$ - both fail at line 2 (3 and 6 rather than 4).

Takeaway Verify a conjectured pattern against **every** given line, and check each component (start, length) separately.

Question 6 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Full method - how and why

Deduction from two premises. Test the strongest-looking option against premise (i) first.

- 1 Suppose n were a multiple of 12. Then $12 = 4 \times 3$ and $12 = 6 \times 2$, so n is a multiple of both 4 and 6.

why: Exploring the tempting case first often resolves the whole question.

- 2 But (i) says a multiple of 4 is not a multiple of 6 - contradiction. So n is **not** a multiple of 12.

why: A supposition leading to a contradiction is ruled out.

- 3 Premise (ii) offers two alternatives and the first is now impossible, so n must be a multiple of 4.

why: Disjunctive syllogism: ' A or B ' plus 'not A ' gives B .

- 4 Then (i) applies: n is a multiple of 4 but not of 6.

why: That is the full deduction.

Where it goes wrong

- Concluding n is a multiple of 12 by picking the first branch of (ii) without testing it.
- Declaring the premises inconsistent - they are perfectly consistent (e.g. $n = 4$ or $n = 8$).
- Concluding nothing, missing that one branch of the 'or' is eliminated.

Takeaway With ' A or B ', eliminate one branch and the other must hold; always test the branch that clashes with the other premise.

Question 7 PRF-15 · Mathematical proof · spec Prf3

Let $u_n = n^2 - n + 41$. Which of the following statements is/are true?

- I u_n is prime for $n = 1, 2, 3, 4, 5$.
- II u_n is prime for all positive integers n .
- III u_{41} is divisible by 41.

Answer: F

$u_1 = 41, u_2 = 43, u_3 = 47, u_4 = 53, u_5 = 61$: all prime (I). $u_{41} = 41^2 - 41 + 41 = 41^2$, divisible by 41 (III) and not prime, so II is false - small cases do not prove a general claim.

Full method - how and why

The classic warning about pattern-spotting: a formula can produce primes for many values and still fail.

- 1 I: compute $u_1 = 41, u_2 = 4 - 2 + 41 = 43, u_3 = 9 - 3 + 41 = 47, u_4 = 16 - 4 + 41 = 53, u_5 = 25 - 5 + 41 = 61$.
All prime. **True.**

why: Straight substitution; all five are primes (none divisible by 2, 3, 5 and each is below 64 or checkable quickly).

- 2 III: $u_{41} = 41^2 - 41 + 41 = 41^2$. Divisible by 41. **True.**

why: The -41 and $+41$ cancel exactly - which is why $n = 41$ is the natural place to look for failure.

- 3 II: since $u_{41} = 41^2 = 1681$ has factor 41, it is not prime. **False.**

why: One counterexample kills the universal claim, however many earlier cases worked.

- 4 So I and III.

why: Match to the list.

Where it goes wrong

- Accepting II because the first several values are prime - this formula is famous precisely because it works for $n = 1$ to 40 and then fails.
- Trying to check u_{41} by multiplying out 1681 instead of spotting the cancellation.

Takeaway Many confirming cases never prove a universal claim; and when hunting for failure, look for the value that makes terms cancel.

Question 8 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I S contains a positive multiple of 5.
- II S contains no positive multiple of 3.
- III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.

Full method - how and why

'Must be true' with a concrete object given. Some claims follow from the given element; others need a counterexample set.

- 1 I: $10 \in S$ and $10 = 2 \times 5$ is a positive multiple of 5. **Must be true.**
why: The element is handed to you; nothing more is needed.
- 2 II: the second condition says every multiple of 3 in S is negative, so none can be positive. **Must be true.**
why: Note this is vacuously fine even if S contains no multiples of 3 at all.
- 3 III: must S contain a negative number? Try $S = \{10\}$: every element is a multiple of 3 or 5 ✓, there are no multiples of 3 to be negative ✓, and $10 \in S$ ✓. No negative number. **Not necessarily true.**
why: A single valid S with no negative element refutes 'must be true'.
- 4 So I and II.
why: Match to the list.

Where it goes wrong

- Reading 'every multiple of 3 in S is negative' as promising that S **has** a multiple of 3 - a universal statement can be true with nothing to apply it to.
- Confusing 'could be true' with 'must be true' for III.

Takeaway A conditional statement about members of a set is vacuously true when no member qualifies; to refute 'must be true', build a legal example that fails it.

Question 9 PRF-19 · Mathematical proof · spec Prf1

A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2$...' At which point does the argument break down compared with the $\sqrt{2}$ proof?

Answer: C

For $\sqrt{2}$ the step $p^2 = 2q^2$ leads to $2k^2 = q^2$, making q even. Here $k^2 = q^2$ carries no parity information, so the contradiction never appears - as expected since $\sqrt{4} = 2$.

Full method - how and why

Diagnosing why a proof technique fails when transplanted. Follow the argument line by line and find the step that no longer bites.

- 1 Suppose $\sqrt{4} = \frac{p}{q}$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even - this much is **fine**.
why: $4q^2$ is certainly even, and an odd number squared is odd, so p must be even.
- 2 Substituting $p = 2k$: $4k^2 = 4q^2$, i.e. $k^2 = q^2$.
why: Here the 4 cancels entirely, because the constant was 4 and not 2.
- 3 Compare with $\sqrt{2}$: there $p^2 = 2q^2$ gives $4k^2 = 2q^2$, i.e. $q^2 = 2k^2$ - a spare factor of 2 forcing q even.
why: That leftover factor of 2 is the whole engine of the classical proof.
- 4 With $\sqrt{4}$ no factor is left over, so $k^2 = q^2$ tells you nothing about the parity of q , and no contradiction ever appears.
why: Correctly so: $\sqrt{4} = 2$ is rational, so no proof of irrationality could possibly succeed.

Where it goes wrong

- Claiming $p^2 = 4q^2$ fails to make p even - it does not fail; that step is valid.
- Claiming the 'lowest terms' assumption is illegitimate - every rational can be written that way.
- Accepting the argument and concluding $\sqrt{4}$ is irrational, when $\sqrt{4} = 2$.

Takeaway The $\sqrt{2}$ proof works because 2 is prime; the leftover prime factor is what forces the second parity. Check where an adapted proof loses its engine.